

## Drill Problems 12

*Author: Jaehoon Song**Release: 2025-08-10 15:10:53-04:00***Purpose and Usage:**

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C O L U M B I A   A C A D E M Y

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Written by Jaehoon Song

1. **Solving for  $2n$**  (10 points)

If  $6n + 40 = 3(2n - 20) + 2n$ , what is the value of  $2n$ ?

**Answer:** Let's solve step by step:

$$6n + 40 = 3(2n - 20) + 2n$$

Distribute the 3:  $6n + 40 = 6n - 60 + 2n$

Combine like terms:  $6n + 40 = 8n - 60$

Subtract  $6n$  from both sides:  $40 = 2n - 60$

Add 60 to both sides:  $100 = 2n$

Therefore,  $2n = 100$

□

2. **Game Store Sales** (10 points)

The game store sells both board games and figurines. Board games sell for \$40 a game and figurines sell for \$15 a figurine. If the store sells 80 total pieces of merchandise and earned \$2625 from those sales, how many board games did the game store sell?

(A) 23

(B) 40

(C) 48

(D) 57

**Answer:** Let  $b$  = number of board games and  $f$  = number of figurines.

We have two equations:  $b + f = 80$  (total pieces)  $40b + 15f = 2625$  (total revenue)

From the first equation:  $f = 80 - b$

Substitute into the second equation:  $40b + 15(80 - b) = 2625$   $40b + 1200 - 15b = 2625$   $25b + 1200 = 2625$   $25b = 1425$   $b = 57$

The store sold 57 board games.

Answer: D

□

3. **Polygon Exterior Angle** (10 points)

The equation  $nA = 360$  shows the relationship between the measure  $A$ , in degrees, of an exterior angle of a regular polygon and  $n$ , the number of sides of the polygon. If a regular polygon's exterior angle is greater than 80 degrees, what is the greatest number of sides it can have?

(A) 4

(B) 5

(C) 6

(D) 8

**Answer:**

□

4. **Scatterplot Inference** (10 points)

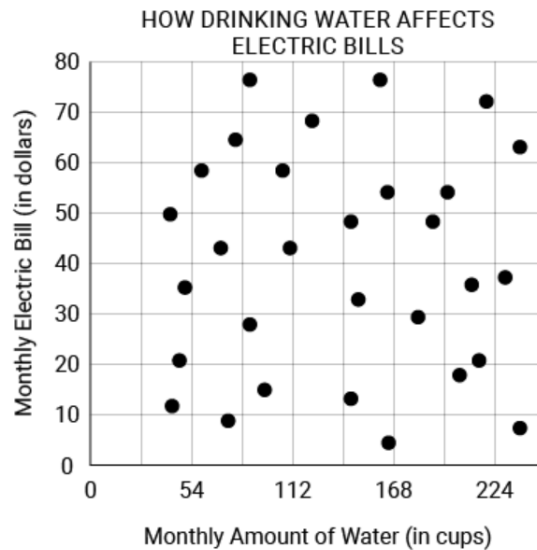


Figure 1: reference attached

A study asked voluntary participants to record the number of cups of water they drink in a month and the electric bill for the month. The results are shown in the scatterplot. What can be inferred from the scatterplot?

- (A) No one drinks less than 27 cups of water in a month.
- (B) The more water one drinks, the higher the electric bill.
- (C) There is no correlation between the monthly amount of water drunk and the electric bill.
- (D) Drinking bottled water explains why some electric bills are lower.

**Answer:**

□

5. **Right Triangle Hypotenuse** (10 points)

A right triangle has legs with lengths of 15 centimeters and 25 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form  $5\sqrt{n}$ , where  $n$  is an integer, what is the value of  $n$ ?

**Answer:**

□

6. Slope of Perpendicular Line (10 points)

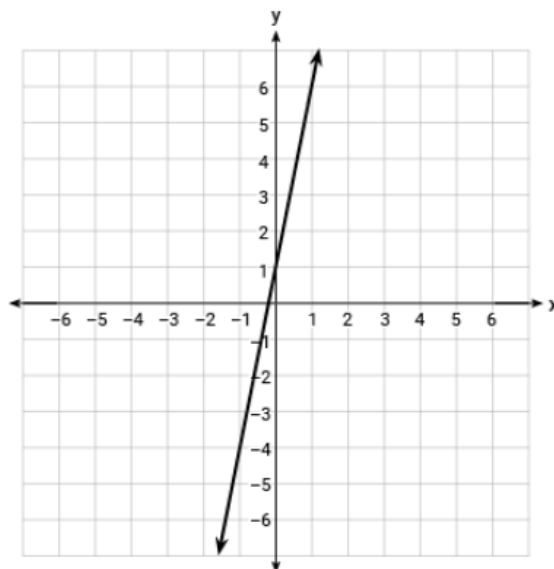


Figure 2: reference attached

The graph of line  $q$  is shown above. Line  $r$  is perpendicular to line  $q$ . What is the slope of line  $r$ ?

- (A)  $-5$
- (B)  $-\frac{1}{5}$
- (C)  $\frac{1}{5}$
- (D)  $5$

**Answer:**

□

7. Perpendicular Linear Functions (10 points)

For the linear function  $g(x)$ ,  $g(2.6) = 11$  and  $g(5) = 3.8$ . When graphed on the coordinate plane, the function  $g(x)$  is perpendicular to the graph of the linear function  $h(x) = mx$ . What is the value of  $m$ ?

- (A)  $-3$
- (B)  $-\frac{1}{3}$
- (C)  $\frac{1}{3}$
- (D)  $3$

**Answer:**

□

8. **Bacteria Population Growth** (10 points)

Harold prepared a sample of bacteria. A strain of virus killed 30 percent of the bacteria, but within a week, the surviving population had grown 47 percent. Which of the following represents the value of the original population in terms of the new population  $p$ ?

- (A)  $\frac{p}{1.17}$
- (B)  $1.17p$
- (C)  $\frac{p}{(0.70)(1.47)}$
- (D)  $(0.70)(1.47)p$

**Answer:**

□

9. **Fruit Fly Population Model** (10 points)

The population of fruit flies in a laboratory study triples every 7 days. The population at the beginning of the study numbered 200. Which equation represents the population of fruit flies,  $y$ , in the study  $t$  weeks after the beginning of the study?

- (A)  $y = 200(3)^t$
- (B)  $y = 3(200)^t$
- (C)  $y = 200(3)^{\frac{t}{7}}$
- (D)  $y = 200^{3t}$

**Answer:**

□

10. **Infinite Solutions in a System** (10 points)

Consider the system:

$$\begin{aligned} 24x + 24y &= -8y + \frac{2}{3} \\ -ky &= 3x - \frac{1}{12} \end{aligned}$$

In the given system of equations,  $k$  is a constant. If the system has infinite solutions, what is the value of  $k$ ?

**Answer:**

□

11. **Oxygen Drop Rate in Ocean** (10 points)

As a probe descends to the Kuril Trench in the Pacific Ocean, it records the amount of dissolved oxygen in the water in parts per million (ppm). At depths below 2,000 meters, the dissolved oxygen drops at a constant rate of  $d$  ppm for every change in depth of 100 meters until it reaches zero. At a depth of 2,530 meters, the dissolved oxygen is measured to be 205 ppm. At a depth of 3,030 meters, the dissolved oxygen is measured to be 65 ppm. What is the value of  $d$ ?

**Answer:**

□

12. **Computer Purchase Budget** (10 points)

Mrs. Brennan wants to buy  $x$  computers for her company and has a budget of  $B$  dollars. If she buys  $x$  computers today, they will cost \$1,000 each, and she will need to raise an additional \$2,000. If she waits two months, the computers will cost \$900 each, and she will have \$400 left over. What is the value of  $B$ ?

- (A) 14,800
- (B) 21,600
- (C) 22,000
- (D) 24,000

**Answer:**



13. **No  $x$ -Intercepts for a Quadratic** (10 points)

Let  $f(x) = x(x - 14) + 70 - k$ , where  $k$  is an integer constant. For which of the following values of  $k$  will the graph of the function have no  $x$ -intercepts?

- (A) 20
- (B) 21
- (C) 25
- (D) 30

**Answer:**



14. **Water Boy's Weekly Hours** (10 points)

A professional football team practices 6 days a week in the preseason. A water boy for the team knows that the players drink about 200 gallons of water each day of preseason practice. If the water tank in the utility room pours water at a rate of 9 gallons per minute, how many hours a week does the water boy spend filling water containers?

- (A)  $\frac{10}{27}$
- (B)  $\frac{20}{9}$
- (C)  $\frac{100}{27}$
- (D)  $\frac{400}{3}$

**Answer:**



15. **Radius of a Circle from Equation** (10 points)

The graph of  $x^2 + 3x + y^2 - y = \frac{13}{2}$  in the  $xy$ -plane is a circle. What is the length of the circle's radius?

**Answer:**



16. **No Solution for a Linear Equation with Parameters** (10 points)

Consider the equation:

$$3(ax + b) = \frac{21}{10}x - \frac{48}{11}$$

In the given equation,  $a$  and  $b$  are constants and  $b > 0$ . The equation has no solution. What is the value of  $a$ ?

**Answer:**

□

17. **Algebraic Expression Equivalence** (10 points)

Which expression is equivalent to  $\frac{1}{\frac{1}{x+5} + \frac{1}{x-2}}$ ?

- (A)  $\frac{2}{2x+3}$
- (B)  $\frac{2x+3}{2}$
- (C)  $\frac{2x+3}{x^2+3x-10}$
- (D)  $\frac{x^2+3x-10}{2x+3}$

**Answer:**

□

18. **Box Plot Data Set Comparison** (10 points)

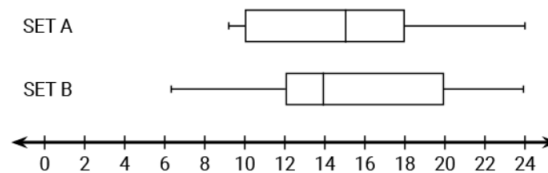


Figure 3: reference attached

The box plot summarizes the values in two different data sets. Based on the box plot, which of the following must be true?

- (A) The mean of set  $A$  is greater than the mean of set  $B$ .
- (B) The range of set  $A$  is greater than the range of set  $B$ .
- (C) The median of set  $A$  is greater than the median of set  $B$ .
- (D) The standard deviation of set  $A$  is greater than the standard deviation of set  $B$ .

**Answer:**

□

19. **Number of Real Solutions to a Quadratic** (10 points)

How many distinct real solutions does the equation  $0.6x^2 - 1.5x + 2.01 = 0$  have?

- (A) Zero
- (B) Exactly one
- (C) Exactly two
- (D) Infinitely many

**Answer:**



20. **Area of Shaded Region in Hexagon** (10 points)

A regular hexagon is inscribed in a rectangle as shown.

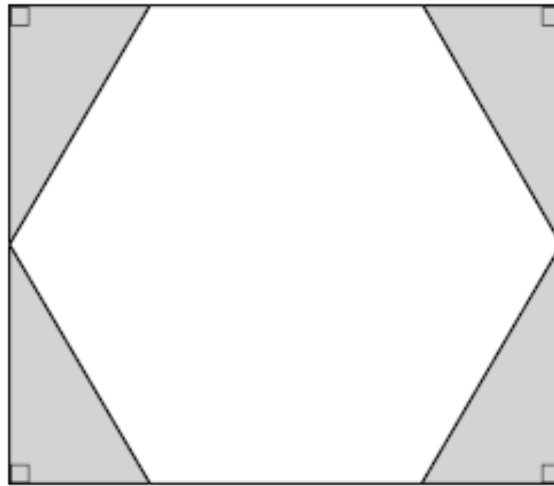


Figure 4: reference attached

The perimeter of the hexagon is  $6a$  units. What is the area of the shaded region, in square units?

- (A)  $\frac{\sqrt{3}}{8}a^2$
- (B)  $\frac{\sqrt{3}}{2}a^2$
- (C)  $\frac{3\sqrt{3}}{2}a^2$
- (D)  $2\sqrt{3}a^2$

**Answer:**





21. **Parameter in Polynomial Factorization** (10 points)

The function  $g(x)$  is defined as  $g(x) = (x^2 + bx + c)(x + a)$ , where  $a, b, c$  are constants. There are two distinct values such that  $g(x) = 0$ . If  $g(x)$  can also be written in the form  $\frac{1}{m}(mx^2 + mbx + b^2)(x + a)$ , what is the value of  $m$ ?

**Note:**  $-a$  must be one of the two distinct answer.

**Answer:**



22. **Product of Function Values** (10 points)

The function  $f$  is defined by the equation  $f(x) = \frac{x}{a-9}$ , where  $a$  is a positive constant and  $a \neq 9$ . If the value of the product of  $f(2a)$  and  $f(18a)$  is equal to  $\frac{9}{100}$ , what is the value of  $a$ ?

**Answer:**



23. **System of Equations for Cookies and Brownies** (10 points)

Angela buys 2 cookies and 3 brownies for a total of \$19. Vicky buys 4 cookies and 1 brownie for a total of \$13. If  $c$  represents the cost of a cookie and  $b$  represents the cost of a brownie, which system of equations can determine the cost of one cookie and one brownie?

(A)

$$c + b = 5$$

$$6c + 4b = 32$$

(B)

$$2c + 3b = 19$$

$$4c + b = 13$$

(C)

$$5c = 19b$$

$$5b = 13c$$

(D)

$$6c = 19$$

$$4b = 13$$

**Answer:** B



24. **Factoring Quadratic Expression** (10 points)

Which expression is equivalent to  $x^2 + 5x - 6$ ?

(A)  $(x + 2)(x - 3)$

(B)  $(x - 2)(x + 3)$

(C)  $(x + 1)(x - 6)$

(D)  $(x - 1)(x + 6)$

**Answer:** D



25. **Equation of a Circle** (10 points)

A circle in the  $xy$ -plane has its center at  $(-3, 7)$ , and the point  $(1, 7)$  lies on the circle. Which equation represents this circle?

- (A)  $(x - 3)^2 + (y + 7)^2 = 4$
- (B)  $(x + 3)^2 + (y - 7)^2 = 4$
- (C)  $(x - 3)^2 + (y + 7)^2 = 16$
- (D)  $(x + 3)^2 + (y - 7)^2 = 16$

**Answer:** D



26. **Parallel Lines and Slope** (10 points)

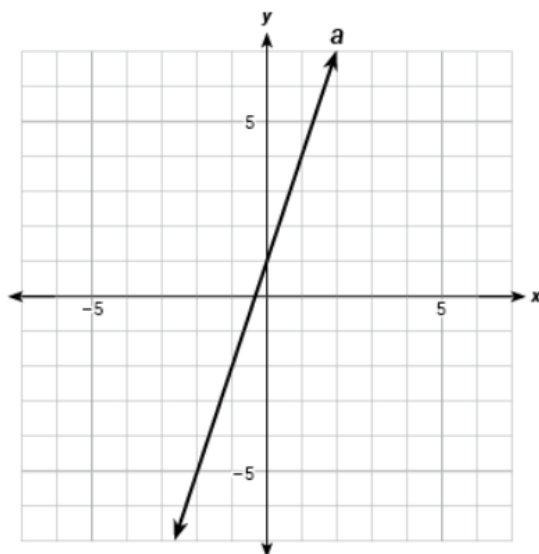


Figure 5: reference attached

The graph of line  $a$  is shown, and the equation of line  $a$  is  $y = 3x + 1$ . Line  $b$  is parallel to line  $a$  and crosses through the points  $(0, 5)$  and  $(k, -7)$ . What is the value of  $k$ ?

**Answer:**



27. **Solution to a Graphed Linear Inequality** (10 points)

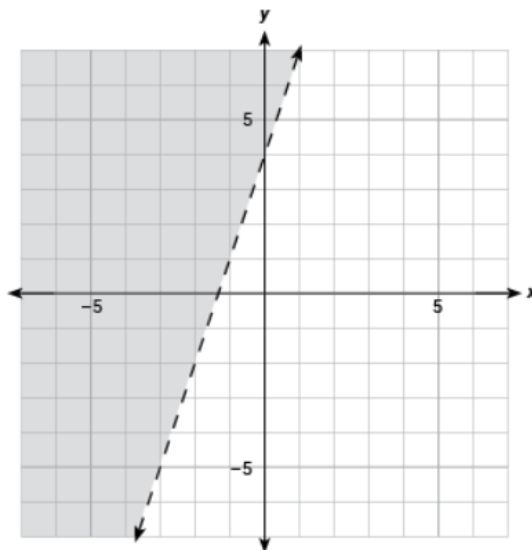


Figure 6: reference attached

A linear inequality is graphed on the  $xy$ -coordinate plane. Which coordinate point is a solution to the graphed inequality?

- (A)  $(-3, -4)$
- (B)  $(-2, -3)$
- (C)  $(0, 4)$
- (D)  $(1, 6)$

**Answer: A**



28. **Effect of Removing a Value from a Data Set** (10 points)

Data set  $X$  : 25, 32, 11, 42, 33, 16, 27, 39, 54

If 54 is removed from the data set, which of the following is true?

- (A) The mean and the median do not change.
- (B) The mean and the median change by the same amount.
- (C) The change to the median is greater than the change to the mean.
- (D) The change to the mean is greater than the change to the median.

**Answer: D**



29. **Quadratic Equation Solution** (10 points)

What is one of the solutions to the equation  $2x^2 - 40x + 182 = 0$ ?

**Answer:**  $x = 7$  or  $x = 13$

□

30. **Number of Solutions to a Linear Equation** (10 points)

How many solutions does the equation  $2(37x + 1) = 37(2x + 1)$  have?

- (A) Exactly one
- (B) Exactly two
- (C) Infinitely many
- (D) Zero

**Answer:** D

□

31. **Scatterplot: Predicted vs Actual Cost** (10 points)

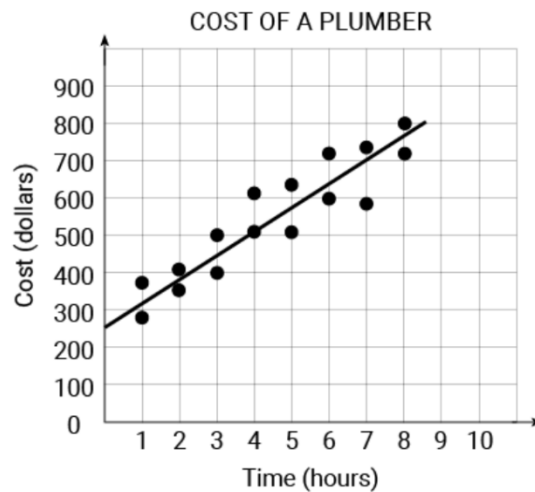


Figure 7: reference attached

The scatterplot shows the comparison for the time and the cost to hire a plumber. At what hour is the predicted cost the closest to an actual cost to hire a plumber?

**Answer:**  $x = 4$

□

32. Perpendicular Lines and  $y$ -Intercept (10 points)

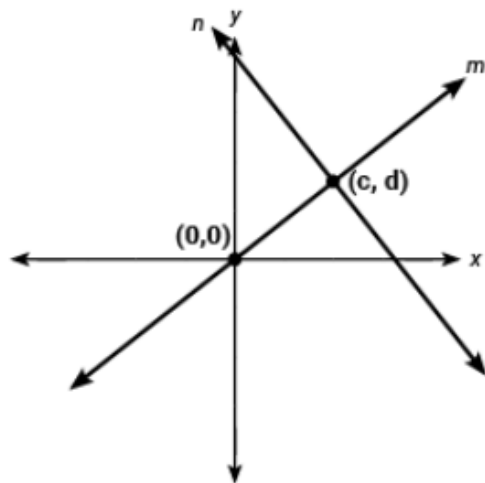


Figure 8: reference attached

Two linear equations  $m$  and  $n$  are graphed on the  $xy$ -coordinate plane, as shown. If line  $m$  is perpendicular to line  $n$ , what is the value of the  $y$ -intercept of line  $n$ ?

- (A)  $d - \frac{c^2}{d}$
- (B)  $d + \frac{c^2}{d}$
- (C)  $d - \frac{d^2}{c}$
- (D)  $d + \frac{d^2}{c}$

**Answer:** B

□

33. Exponential Growth of Fungi (10 points)

In certain conditions, filamentous fungi will grow exponentially, tripling in size every day. If, on day 1, the fungi are each 4 mm in length, which of the functions below represents the length on the  $n^{\text{th}}$  day?

- (A)  $y = 3(4)^n$
- (B)  $y = 3(4)^{n-1}$
- (C)  $y = 4(3)^n$
- (D)  $y = 4(3)^{n-1}$

**Answer:** D

□

34. **Quadratic Equation with No Real Solutions** (10 points)

In the equation  $2x^2 - 27x + d = 0$ ,  $d$  is a positive integer constant. The equation has no real solutions. What is the smallest possible value for  $d$ ?

- (A) 91
- (B) 92
- (C) 93
- (D) 94

**Answer:** B



35. **Solving for  $n$  in Terms of  $m$  and  $p$**  (10 points)

The equation  $m^6 = (n + 5p)^3$  relates the variables  $m$ ,  $n$ , and  $p$  for all real numbers  $> 0$ . Which equation correctly expresses  $n$  in terms of  $m$  and  $p$ ?

- (A)  $n = \sqrt[3]{m^3 - 5p}$
- (B)  $n = m^2 - \sqrt[3]{5p}$
- (C)  $n = m^2 - 5p$
- (D)  $n = m^3 - 5p$

**Answer:** C



36. **Trigonometry: Solving for  $x$**  (10 points)

In the right triangle  $ABC$ , angle  $C$  is a right angle. If  $\cos(A) = \frac{x-k}{x+k}$ , where  $k$  is a constant and  $x$  is the side length of  $BC$ , what is the value of  $x$  in terms of  $k$ ?

- (A)  $2k$
- (B)  $4k$
- (C)  $2k^2$
- (D)  $4k^2$

**Answer:**



37. **Shirt Order Price Calculation** (10 points)

| Number of shirts | Cost of Shirts | Delivery Cost | Tax     | Total Cost |
|------------------|----------------|---------------|---------|------------|
| 6                | \$210.00       | \$25.20       | \$23.52 | \$258.72   |
| 15               | \$525.00       | \$63.00       | \$58.80 | \$646.80   |
| 21               | \$735.00       | \$88.20       | \$82.32 | \$905.52   |

Figure 9: reference attached

The table above shows the price for ordering different quantities of shirts. A delivery fee is calculated as a percentage of the base cost of the shirts. Then, a tax is applied to the combined cost of the shirts and the delivery. If  $p$  is the percentage added for delivery and  $q$  is the percentage added for taxes, what is the value of  $p - q$ ?

- (A) 0.8
- (B) 2
- (C) 6.67
- (D) 23.2

**Answer:**



38. **Piecewise Soda Pricing** (10 points)

A cruise ship charges \$4 per soda for the first 19 sodas and then charges \$2.50 for each additional soda. If  $c(s)$  is the total cost, in dollars, for  $s$  sodas and  $s > 19$ , what is the equation of  $c(s)$ ?

- (A)  $c(s) = 2.5s + 19$
- (B)  $c(s) = 2.5s + 28.5$
- (C)  $c(s) = 2.5s + 76$
- (D)  $c(s) = 6.5s + 47.5$

**Answer:** B



39. Exponential Function and Translation (10 points)

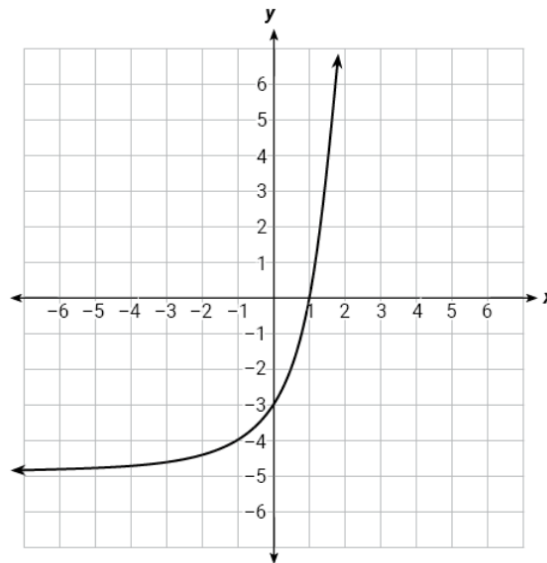


Figure 10: reference attached

The exponential function  $f(x)$  is graphed on the  $xy$ -plane. The function  $g(x)$  (not shown) is defined such that  $g(x) = f(x - 1) + k$ . If  $g(0) = 1$ , what is the value of  $k$ ?

- (A)  $-5$
- (B)  $-4$
- (C)  $4$
- (D)  $5$

**Answer:** D

□

40. Intersection of Lines and Triangle Area (10 points)

Line  $A : y = ax + b$  and line  $B : y = cx + d$ , where  $a, b, c, d$  are constants and  $b > d$ , intersect in the first quadrant of the  $xy$ -coordinate plane at the point  $(j, k)$ . A triangle is formed from the  $y$ -intercepts of the lines and the point of intersection, and the base is formed by the distance between the  $y$ -intercepts of the lines. If the base and height of the triangle are equal, what is the value of  $c - a$ ?

**Answer:**

□



41. **Volume of a Similar Rectangular Pyramid** (10 points)

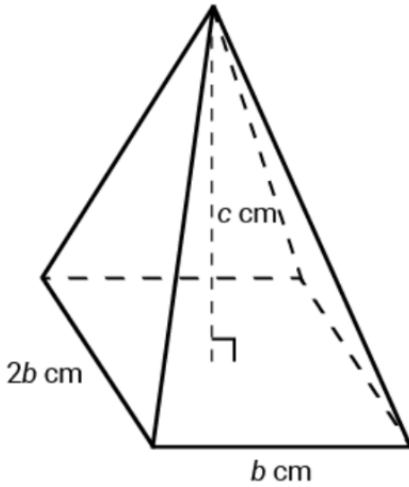


Figure 11: reference attached

A rectangular pyramid has a base length measuring  $b$  centimeters, a base width measuring  $2b$  centimeters, and a height measuring  $c$  centimeters, as shown. A similar rectangular pyramid has side lengths that are all  $\frac{2}{3}$  the side lengths of the given rectangular pyramid. What is the volume of the smaller rectangular pyramid?

- (A)  $\frac{2b^2c}{3}$
- (B)  $\frac{4b^2c}{9}$
- (C)  $\frac{8b^2c}{27}$
- (D)  $\frac{16b^2c}{81}$

**Answer:**



42. **Quadratic Equation with Intercepts and Vertex** (10 points)

A quadratic equation intercepts the  $x$ -axis at two different points,  $(0,0)$  and  $(2j,0)$ , where  $j > 0$ . The distance between the  $x$ -intercepts is equal to the shortest distance from the  $x$ -axis to the vertex, and the vertex is in the first quadrant. Which of the following represents the quadratic equation?

- (A)  $y = \frac{2}{3j}(x)(x - 2j)$
- (B)  $y = -\frac{2}{3j}(x)(x - 2j)$
- (C)  $y = -\frac{j}{2}(x)(x - 2j)$
- (D)  $y = -\frac{2}{j}(x)(x - 2j)$

**Answer:**



43. **Quadratic Equation with Product of Solutions** (10 points)

For the quadratic equation  $kx^2 - 62x + 288 = 0$ , if the product of the solutions is  $-72$ , what is the value of  $k$ ?

**Answer:**  $-4$

□

44. **Ratio in Inscribed Triangle** (10 points)

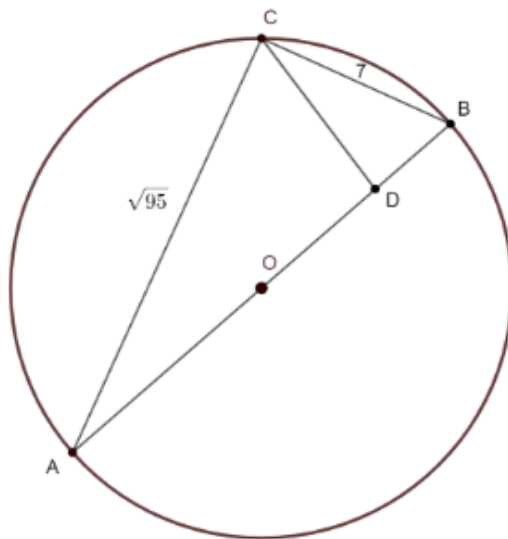


Figure 12: reference attached

Circle  $O$  has its center at point  $O$ , as shown. Points  $A$ ,  $B$ , and  $C$  lie on Circle  $O$  and  $AB$  is the diameter. Triangle  $ACB$  is inscribed in the circle, and altitude  $CD$  is shown. The length of  $CB = 7$  centimeters and  $AC = \sqrt{95}$  centimeters. Which of the following is closest to the ratio of  $CB$  to  $CD$ ?

- (A)  $1.7 : 1$
- (B)  $1.4 : 1$
- (C)  $1.2 : 1$
- (D)  $1.1 : 1$

**Answer:**

□